SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Campus Club and Event Management System

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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements of the **Campus Club and Event Management System**.

The purpose of this system is to manage campus clubs, events, registrations, volunteers, and event budgets in a centralized and organized manner.

The system provides role-based access for different users such as **Admin, Club Head, Event Coordinator, Student, and Volunteer**.

1.2 Scope

The system provides the following functionalities:

• Students can view events, register for events, make payments for paid events, download certificates, and give feedback.

• Club Heads can create clubs, manage team members, create events, and propose budgets.

• Event Coordinators can manage registrations, assign volunteers, verify attendance, record expenses, and generate certificates.

• Volunteers can view assigned tasks and update task status.

• Admin can approve clubs, approve events, approve budgets, manage users, publish announcements, and generate reports.

1.3 Definitions, Acronyms, Abbreviations

SRS – Software Requirement Specification

UI – User Interface

DFD – Data Flow Diagram

DB – Database

Admin – System Administrator

Token – Unique code generated after event registration used for attendance verification

1.4 References

Project Feasibility Report

Use Case Diagram of Campus Club and Event Management System

Data Flow Diagrams (Level 0, Level 1, Level 2)

MySQL Documentation

HTML, CSS, JavaScript Documentation

Software Engineering IEEE SRS Standard

1.5 Overall Description

The Campus Club and Event Management System is a **web-based platform** designed to manage campus clubs and events efficiently.

It integrates students, club heads, event coordinators, and administration under a single centralized system.

The system handles:

• Club creation and approval
• Event planning and management
• Event registration and payments
• Volunteer task management
• Attendance verification using tokens
• Certificate generation
• Event reporting and feedback

2. Overall Description

2.1 Product Perspective

The Campus Club and Event Management System is a **web-based application** that follows a **client-server architecture**.

The system uses a centralized database and role-based access control to ensure security and proper management of club activities and events.

2.2 Product Functions

Admin

• Manage Users
• Approve Club
• Approve Event
• Approve Budget
• Publish Announcements
• View Reports

Club Head

- Submit Club Details
- Create Event
- Propose Budget
- Manage Team

Event Coordinator

- Manage Registrations
- Assign Volunteers
- Verify Attendance
- Record Expenses
- Generate Certificates

Student

- View Events
- Register for Event
- Make Payment
- Generate Token
- Download Certificate
- Give Feedback

Volunteer

- View Assigned Tasks
- Update Task Status

2.3 User Characteristics

Admin

Has technical knowledge and manages the system operations.

Club Head

Responsible for managing club activities and organizing events.

Event Coordinator

Handles event execution and manages volunteers.

Student

Basic computer knowledge and internet access.

Volunteer

Participates in event management tasks.

2.4 User Constraints

- Internet connection required
- Users must have valid login credentials
- Only approved clubs can organize events
- Budget must be approved before event execution
- Only registered students can participate in events

2.5 Assumptions and Dependencies

- Users have valid login credentials
- Server must be available for system access
- Database connectivity must be active

• Event information is provided by club heads

• Students provide valid registration details

2.6 Apportioning Requirements

Future enhancements may include:

• Mobile application for event management

• Integration with college portal

• AI-based event recommendation system

• Automated certificate verification using QR codes

3. Specific Requirements

3.1 Interface Requirements

3.1.1 External Interface

Web browser interface (Chrome, Firefox, Edge)

Responsive web UI

3.1.2 Hardware Interface

Server system

Client device (Laptop / Desktop / Mobile)

Internet connectivity

3.1.3 Software Interface

Web server (Apache / Node.js)

Backend technologies (Java / PHP / Python)

Database (MySQL)

3.1.4 Communication Interface

HTTP / HTTPS protocol

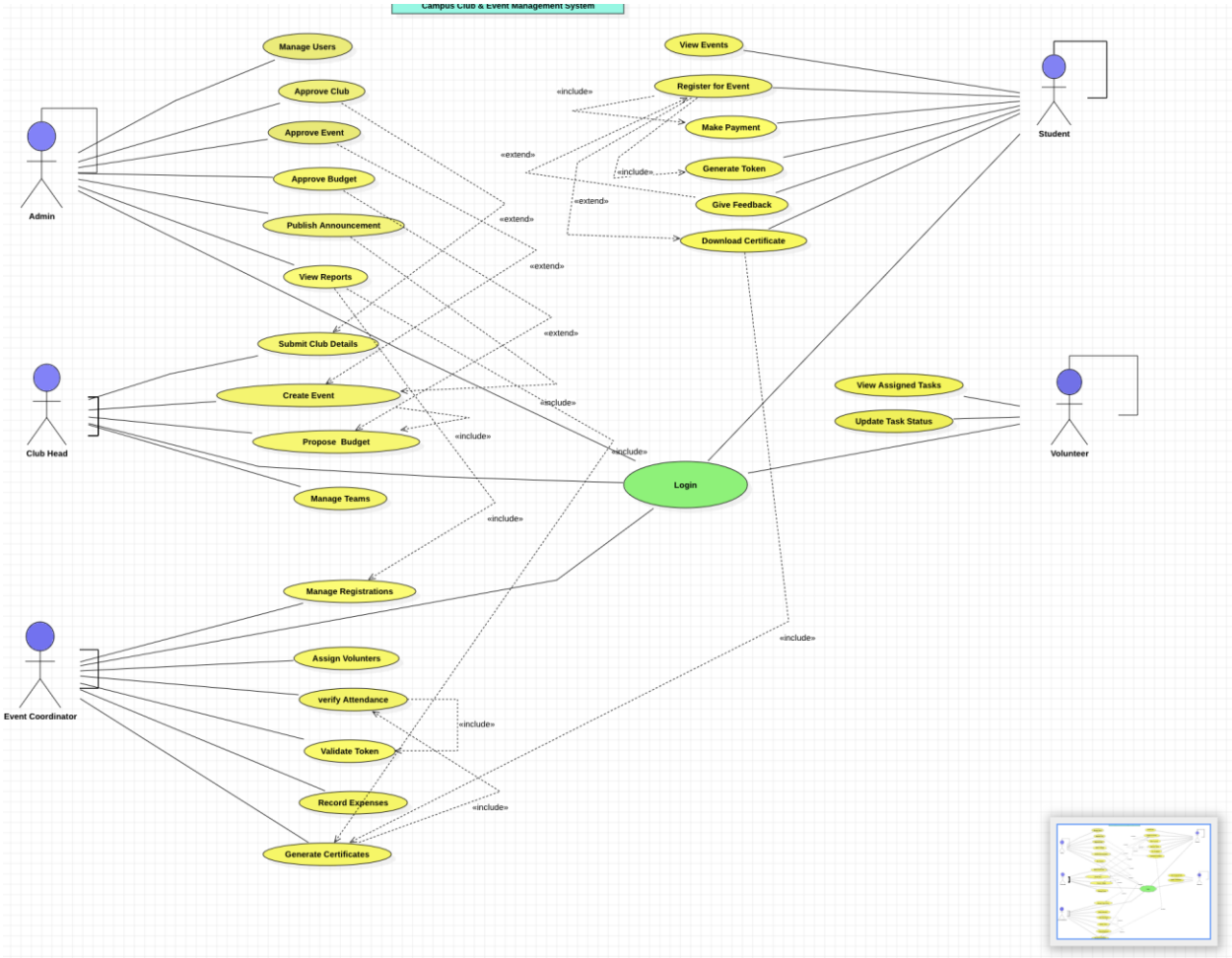
Secure login authentication

Email notifications for event updates

3.2 Functional Requirements

3.2.1 Use Case Model / Information Flow

Use Case Diagram



3.2.2 Use Case Specifications

Insert the **use case tables you already prepared** for:

- Manage Users
- Approve Club
- Approve Event

- Approve Budget
- Publish Announcements
- View Reports
- Submit Club Details
- Create Event
- Propose Budget
- Manage Team
- Manage Registrations
- Assign Volunteers
- Verify Attendance
- Record Expenses
- Generate Certificates
- View Events
- Register for Event
- Make Payment
- Generate Token
- Download Certificate
- Give Feedback
- View Assigned Tasks
- Update Task Status

Field	Details
Use Case ID	UC-02
Use Case Name	Approve Club
Actors	Admin
Description	Allows admin to review and approve or reject club registration requests submitted by club heads.
Preconditions	Club details must be submitted by the club head.
Post Conditions	Club status updated as approved or rejected.
Normal Flow	1. Admin logs in. 2. Opens club requests section. 3. Reviews club details. 4. Approves or rejects the club. 5. System updates club status.

Alternate Flows	A1: If club details are incomplete → request rejected.
Exceptions	System error while updating club information.
Priority	High
Frequency	Occasionally
Business Rules	Only admin can approve or reject clubs.
Assumptions	Club head submitted valid club information.

Field	Details
Use Case ID	UC-03
Use Case Name	Approve Event
Actors	Admin
Description	Allows admin to review and approve or reject events created by club heads.
Preconditions	Event must be created and submitted by the club head.
Post Conditions	Event status updated and event becomes available to students if approved.
Normal Flow	1. Admin logs in. 2. Opens event requests. 3. Reviews event details. 4. Approves or rejects event. 5. System updates event status.
Alternate Flows	A1: If event details are invalid → event rejected.
Exceptions	System error while updating event status.
Priority	High
Frequency	Multiple times
Business Rules	Events must be approved by admin before publishing.
Assumptions	Event request already stored in system.

Field	Details
Use Case ID	UC-04

Use Case Name	Approve Budget
Actors	Admin
Description	Allows admin to review and approve or reject the budget proposed by the club head for an event.
Preconditions	Event must be created and budget proposal must be submitted by the club head.
Post Conditions	Budget status updated as approved or rejected.
Normal Flow	1. Admin logs in. 2. Opens budget proposals section. 3. Reviews proposed budget details. 4. Approves or rejects the budget. 5. System updates budget status.
Alternate Flows	A1: If budget details are incorrect → admin rejects proposal.
Exceptions	System error while updating budget information.
Priority	High
Frequency	As required
Business Rules	Budget must be approved before event execution.
Assumptions	Club head submitted valid budget details.

Field	Details
Use Case ID	UC-05
Use Case Name	Publish Announcements
Actors	Admin
Description	Allows admin to publish announcements related to events, clubs, or important campus information.
Preconditions	Admin must be logged into the system.
Post Conditions	Announcement becomes visible to all users in the system.
Normal Flow	1. Admin logs in. 2. Opens announcement section. 3. Enters announcement details. 4. Clicks publish. 5. System stores and displays the announcement.

Alternate Flows	A1: If announcement details are incomplete → system prompts correction.
Exceptions	System error while publishing announcement.
Priority	Medium
Frequency	Multiple times
Business Rules	Only admin can publish announcements.
Assumptions	Announcement information provided by admin is valid.

Field	Details
Use Case ID	UC-06
Use Case Name	View Reports
Actors	Admin
Description	Allows admin to view reports related to events, participation, budgets, and other system activities.
Preconditions	Admin must be logged into the system.
Post Conditions	Reports are displayed on the screen for analysis.
Normal Flow	1. Admin logs in. 2. Opens reports section. 3. Selects report type (event report, participation report, budget report). 4. System retrieves data. 5. Report displayed to admin.
Alternate Flows	A1: If no report data is available → system shows message “No data available”.
Exceptions	System error while generating report.
Priority	Medium
Frequency	As required
Business Rules	Only authorized admin can access system reports.
Assumptions	Event and participation data are stored in the system database.

Field	Details
Use Case ID	UC-07
Use Case Name	Submit Club Details
Actors	Club Head

Description	Allows club head to submit club details for registration and approval by admin.
Preconditions	Club head must be registered and logged into the system.
Post Conditions	Club request submitted and waiting for admin approval.
Normal Flow	1. Club head logs in. 2. Opens club registration form. 3. Enters club details (club name, description, members). 4. Submits club request. 5. System stores club information.
Alternate Flows	A1: If required details are missing → system prompts user to fill missing fields.
Exceptions	System error while saving club details.
Priority	High
Frequency	Once per club
Business Rules	Club must be approved by admin before creating events.
Assumptions	Club head has valid system access.

Field	Details
Use Case ID	UC-08
Use Case Name	Create Event
Actors	Club Head
Description	Allows the club head to create a new event for the club including event details such as name, date, venue, and description.
Preconditions	Club must be approved and club head must be logged into the system.
Post Conditions	Event request submitted and waiting for admin approval.
Normal Flow	1. Club head logs in. 2. Opens event creation page. 3. Enters event details (event name, date, venue, description). 4. Submits event request. 5. System stores event details for admin approval.
Alternate Flows	A1: If event details are incomplete → system prompts user to complete the form.

Exceptions	System error while storing event information.
Priority	High
Frequency	Multiple times
Business Rules	Only club heads can create events for their clubs.
Assumptions	Club is registered and approved by admin.

Field	Details
Use Case ID	UC-09
Use Case Name	Propose Budget
Actors	Club Head
Description	Allows the club head to propose a budget for the event including estimated expenses.
Preconditions	Event must be created and club head must be logged into the system.
Post Conditions	Budget proposal submitted to admin for approval.
Normal Flow	1. Club head selects event. 2. Enters estimated budget details. 3. Submits budget proposal. 4. System stores budget request. 5. Admin reviews the proposal.
Alternate Flows	A1: If budget details are incomplete → system prompts user to enter required information.
Exceptions	System error while saving budget information.
Priority	High
Frequency	Once per event
Business Rules	Budget must be approved by admin before event execution.
Assumptions	Event details are already stored in the system.

Field	Details
Use Case ID	UC-10
Use Case Name	Manage Team
Actors	Club Head

Description	Allows the club head to manage club members, view team details, participation records, and club performance information.
Preconditions	Club must be approved and club head must be logged into the system.
Post Conditions	Team information is updated and displayed in the system.
Normal Flow	1. Club head logs in to the system. 2. Opens the team management section. 3. Views club members and their roles. 4. Updates or manages team details. 5. System saves updated team information.
Alternate Flows	A1: If member information is incorrect, the system prompts correction.
Exceptions	System error while updating team information.
Priority	Medium
Frequency	Multiple times
Business Rules	Only club head can manage team members of the club.
Assumptions	Club members are already registered in the system.

Field	Details
Use Case ID	UC-11
Use Case Name	Manage Registrations
Actors	Event Coordinator
Description	Allows the event coordinator to manage student registrations for events and view the list of registered participants.
Preconditions	Event must be created and students must be registered for the event.
Post Conditions	Registration records are updated and displayed.
Normal Flow	1. Event coordinator logs in to the system. 2. Opens event registration section. 3. Selects the event. 4. Views the list of registered students.

	5. Updates or manages registration details if necessary.
Alternate Flows	A1: If no students are registered, the system displays a message “No registrations available”.
Exceptions	System error while retrieving registration information.
Priority	High
Frequency	Multiple times
Business Rules	Only event coordinators can manage event registrations.
Assumptions	Students have already registered for the event.

Field	Details
Use Case ID	UC-12
Use Case Name	Assign Volunteers
Actors	Event Coordinator
Description	Allows the event coordinator to assign volunteers to specific tasks required for organizing the event.
Preconditions	Volunteers must be registered and event must exist.
Post Conditions	Volunteers are assigned tasks for the event.
Normal Flow	1. Event coordinator logs in to the system. 2. Opens volunteer management section. 3. Selects the event. 4. Chooses volunteers from the list. 5. Assigns tasks to volunteers. 6. System stores the task assignments.
Alternate Flows	A1: If no volunteers are available, the system displays a notification.
Exceptions	System error while assigning volunteers.
Priority	Medium
Frequency	Once per event
Business Rules	Only event coordinators can assign volunteers to event tasks.
Assumptions	Volunteers are already registered in the system.

Field	Details
Use Case ID	UC-13
Use Case Name	Verify Attendance
Actors	Event Coordinator
Description	Allows the event coordinator to verify and record the attendance of students who participate in the event using event tokens or QR codes.
Preconditions	Student must be registered for the event and must have a valid event token.
Post Conditions	Attendance of the student is successfully recorded in the system.
Normal Flow	1. Event coordinator logs in to the system. 2. Opens the attendance verification section. 3. Scans or enters the student token. 4. System validates the token. 5. Attendance is recorded.
Alternate Flows	A1: If token is invalid, system displays an error message.
Exceptions	System error while validating token or saving attendance.
Priority	High
Frequency	Multiple times during the event
Business Rules	Only registered students with valid tokens can mark attendance.
Assumptions	Students have received tokens after registration.

Field	Details
Use Case ID	UC-13
Use Case Name	Verify Attendance
Actors	Event Coordinator
Description	Allows the event coordinator to verify and record the attendance of students who participate in the event using event tokens or QR codes.

Preconditions	Student must be registered for the event and must have a valid event token.
Post Conditions	Attendance of the student is successfully recorded in the system.
Normal Flow	1. Event coordinator logs in to the system. 2. Opens the attendance verification section. 3. Scans or enters the student token. 4. System validates the token. 5. Attendance is recorded.
Alternate Flows	A1: If token is invalid, system displays an error message.
Exceptions	System error while validating token or saving attendance.
Priority	High
Frequency	Multiple times during the event
Business Rules	Only registered students with valid tokens can mark attendance.
Assumptions	Students have received tokens after registration.

Field	Details
Use Case ID	UC-15
Use Case Name	Generate Certificates
Actors	Event Coordinator
Description	Allows the event coordinator to generate certificates for students who participated in the event and for winners.
Preconditions	Event must be completed and participant attendance must be verified.
Post Conditions	Certificates are generated and stored in the system for students to download.
Normal Flow	1. Event coordinator logs in to the system. 2. Opens certificate generation section. 3. Selects the completed event. 4. System retrieves the list of participants. 5. Certificates are generated for participants and winners. 6. Certificates are stored in the system.

Alternate Flows	A1: If participant list is empty, system displays message “No participants found”.
Exceptions	System error while generating certificates.
Priority	Medium
Frequency	Once after each event
Business Rules	Only participants and winners receive certificates.
Assumptions	Attendance records are already available in the system.

Field	Details
Use Case ID	UC-16
Use Case Name	View Events
Actors	Student
Description	Allows students to view the list of available campus club events along with event details.
Preconditions	Events must be created and approved by admin.
Post Conditions	Event details are displayed to the student.
Normal Flow	1. Student logs in to the system. 2. Opens the events section. 3. System displays list of available events. 4. Student selects an event to view details.
Alternate Flows	A1: If no events are available, system displays message “No events available”.
Exceptions	System error while retrieving event information.
Priority	Medium
Frequency	Multiple times
Business Rules	Only approved events are visible to students.
Assumptions	Event details are already stored in the system database.

Field	Details
Use Case ID	UC-17
Use Case Name	Register for Event

Actors	Student
Description	Allows students to register for campus events organized by clubs.
Preconditions	Student must be logged into the system and the event must be available for registration.
Post Conditions	Student registration is successfully recorded in the system.
Normal Flow	1. Student logs in to the system. 2. Opens the events section. 3. Selects an event. 4. Clicks register. 5. System records the registration. 6. If the event is paid, payment process begins.
Alternate Flows	A1: If event capacity is full, registration is rejected.
Exceptions	System error while processing registration.
Priority	High
Frequency	Multiple times
Business Rules	A student can register only once for an event.
Assumptions	Event is approved and open for registration.

Field	Details
Use Case ID	UC-18
Use Case Name	Make Payment
Actors	Student
Description	Allows students to make payment for paid events during the registration process.
Preconditions	Student must be registered for the event and the event must require payment.
Post Conditions	Payment is successfully completed and recorded in the system.
Normal Flow	1. Student selects payment option. 2. Enters payment details. 3. Payment gateway processes the payment.

	4. System confirms successful payment. 5. Registration is completed.
Alternate Flows	A1: If payment fails, student is asked to retry payment.
Exceptions	Payment gateway error.
Priority	High
Frequency	Occasional
Business Rules	Payment must be completed for paid events before registration confirmation.
Assumptions	Payment gateway is available and functional.

Field	Details
Use Case ID	UC-18
Use Case Name	Make Payment
Actors	Student
Description	Allows students to make payment for paid events during the event registration process.
Preconditions	Student must be logged into the system and the selected event must require payment.
Post Conditions	Payment is successfully completed and recorded in the system.
Normal Flow	1. Student selects the event requiring payment. 2. Student proceeds to payment page. 3. Student enters payment details. 4. Payment gateway processes the payment. 5. System confirms successful payment and completes registration.
Alternate Flows	A1: If payment fails, the system asks the student to retry the payment.
Exceptions	Payment gateway error or network failure.
Priority	High
Frequency	Occasional
Business Rules	Payment must be completed for paid events before event registration is confirmed.

Assumptions	Payment gateway service is available and working properly.

Field	Details
Use Case ID	UC-19
Use Case Name	Generate Token
Actors	System
Description	The system generates a unique token or QR code for students after successful event registration or payment.
Preconditions	Student must be successfully registered for the event.
Post Conditions	Unique token generated and stored in the system for attendance verification.
Normal Flow	1. Student completes event registration. 2. System generates a unique token or QR code. 3. Token is linked to the student and event. 4. Token is displayed to the student and stored in the system database.
Alternate Flows	A1: If token generation fails, the system attempts to regenerate the token.
Exceptions	System error while generating the token.
Priority	Medium
Frequency	Once per event registration
Business Rules	Each registered student receives a unique token for event entry and attendance verification.
Assumptions	Event registration is successfully completed before token generation.

Field	Details
Use Case ID	UC-20
Use Case Name	Download Certificate
Actors	Student
Description	Allows students to download participation or winner certificates after completion of the event.

Preconditions	Event must be completed and certificates must be generated by the event coordinator.
Post Conditions	Certificate is successfully downloaded by the student.
Normal Flow	1. Student logs in to the system. 2. Opens the certificates section. 3. Selects the event certificate. 4. Clicks download. 5. System downloads the certificate file.
Alternate Flows	A1: If certificate is not available, system displays message “Certificate not available”.
Exceptions	System error while downloading certificate.
Priority	Medium
Frequency	Once per event
Business Rules	Only registered participants or winners can download certificates.
Assumptions	Certificates are generated and stored in the system.

Field	Details
Use Case ID	UC-21
Use Case Name	Give Feedback
Actors	Student
Description	Allows students to provide feedback or suggestions about events they attended.
Preconditions	Student must have participated in the event.
Post Conditions	Feedback is successfully stored in the system.
Normal Flow	1. Student logs in to the system. 2. Opens feedback section. 3. Selects the event. 4. Enters feedback comments and rating. 5. Submits feedback. 6. System stores the feedback.
Alternate Flows	A1: If feedback form is incomplete, system prompts student to enter required details.

Exceptions	System error while saving feedback.
Priority	Medium
Frequency	Once per event
Business Rules	Only students who attended the event can submit feedback.
Assumptions	Event participation data is available in the system.

3.2.3 Analysis Classes / Data Dictionary

Class	Attributes
Admin	adminID, name, email, password
Club	clubID, clubName, description
Event	eventID, eventName, date, venue
Student	studentID, name, email, department
Volunteer	volunteerID, name, assignedTask
Registration	registrationID, studentID, eventID
Budget	budgetID, amount, approvalStatus
Certificate	certificateID, studentID, eventID

3.3 Performance Requirements

- System should support **500+ concurrent users**
- Response time should be **less than 3 seconds**
- Event registration processing should complete within **5 seconds**

3.4 Logical Database Requirements

Relational database (MySQL)

Tables include:

- Admin
- Clubs
- Events
- Students
- Volunteers
- Registrations
- Budgets
- Certificates
- Feedback

3.5 Design Constraints

- Web-based architecture

- Role-based authentication system

- Must follow institutional event management policies

- Data privacy regulations must be maintained

3.6 Software System Attributes

3.6.1 Reliability

- System uptime should be **95% or higher**.
- Backup and recovery mechanism must be implemented.

3.6.2 Availability

System should be available **24/7 except during maintenance**.

3.6.3 Security

- Password encryption
- Role-based access control
- Secure HTTPS communication
- Protection against SQL injection

3.6.4 Maintainability

- Modular system design
- Easy updates for event rules
- Scalable database structure

3.6.5 Portability

- Compatible with Windows and Linux
- Browser-independent
- Can be deployed on cloud platforms

4. Supporting Information

The supporting information for the Campus Club and Event Management System includes:

- Feasibility Report
- Use Case Diagram
- Data Flow Diagrams (Level 0, Level 1, Level 2)
- Use Case Specifications
- IEEE SRS standard template